WEAKLY ASSOCIATED PRIMES CAN BE TESTED AFTER LOCALISING AT THE PRIME

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Allowed standard prerequisites (editorial).

- Weakly associated primes via minimal primes of annihilators. Used in the counted proof at lines 16179-16181. Produces an element whose annihilator has the given prime as a minimal prime over it.
- Exactness of localisation. Used in the counted proof at line 16181. Identifies the annihilator after localising with the localisation of the annihilator.

Lemma 0.1. *Let R be a ring. Let M be an R-module. Let $\mathfrak{p}$ be a prime of R. The following are equivalent:*

(1) *$\mathfrak{p}$ is weakly associated to M,*
(2) *$\mathfrak{p}R_\mathfrak{p}$ is weakly associated to $M_\mathfrak{p}$, and*
(3) *$M_\mathfrak{p}$ contains an element whose annihilator has radical equal to $\mathfrak{p}R_\mathfrak{p}$.*

Proof. Assume (1). Then there exists an element $m \in M$ such that $\mathfrak{p}$ is minimal among the primes containing the annihilator $I = \{x \in R \mid xm = 0\}$ of m. As localization is exact, the annihilator of m in $M_\mathfrak{p}$ is $I_\mathfrak{p}$. Hence $\mathfrak{p}R_\mathfrak{p}$ is a minimal prime of $R_\mathfrak{p}$ containing the annihilator $I_\mathfrak{p}$ of m in $M_\mathfrak{p}$. This implies (2) holds, and also (3) as it implies that $\sqrt{I_\mathfrak{p}} = \mathfrak{p}R_\mathfrak{p}$.

Applying the implication (1) $\Rightarrow$ (3) to $M_\mathfrak{p}$ over $R_\mathfrak{p}$ we see that (2) $\Rightarrow$ (3).

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Finally, assume (3). This means there exists an element $m/f \in M_{\mathfrak{p}}$ whose annihilator has radical equal to $\mathfrak{p}R_{\mathfrak{p}}$. Then the annihilator $I = \{x \in R \mid xm = 0\}$ of m in M is such that $\sqrt{I_{\mathfrak{p}}} = \mathfrak{p}R_{\mathfrak{p}}$. Clearly this means that $\mathfrak{p}$ contains I and is minimal among the primes containing I, i.e., (1) holds. $\square$